Final list of computing services for Fixyachips Global Solutions:

Fixyachips: System Architecture & Permissions Checklist

1. Storage & Input Layer

- **Amazon S3 (Simple Storage Service)**
    - **Role**: Acts as the "Master Vault" for 4K raw footage, the "Streaming Warehouse" for transcoded HLS files, and the "Library" for AI-generated PDF manuals.
    - **Key Permissions**: `s3:Get*`, `s3:Put*`, and `s3:List*` for secure file movement.
    - **Feature**: **S3 Transfer Acceleration** is enabled to ensure your 4K uploads from any location are up to 300% faster.

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2. The Video Production "Factory"

- **AWS Elemental MediaConvert**
    - **Role**: The "Video Smith" that turns 4K masters into HLS streams for global device compatibility (360p to 1080p).
    - **Logo Burn-In**: Automatically merges your brand watermark into the video pixels during transcoding.
    - **Multi-DRM Security**: Encrypts videos at the hardware level (Apple FairPlay/Google Widevine) to ensure screen-recording software only captures a black screen.
    - **Key Permissions**: `mediaconvert:CreateJob` and full S3 read/write access.

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3. The Identity & Security "Vault"

- **Amazon Cognito**
    - **Role**: Your "Front Desk" for student authentication, handling MFA (Multi-Factor Authentication), sign-ups, and password resets.
    - **Key Permissions**: `cognito-idp:*` to manage the student database securely.
- **Amazon CloudFront**
    - **Role**: The "Global Delivery Engine" that serves content from edge locations closest to the student.
    - **Signed Cookies**: Provides a secure "hall pass" for students to stream entire HLS video folders seamlessly.

    - **Signed URLs**: Used for individual, expiring links to premium PDF manual downloads.
- **AWS WAF (Web Application Firewall)**
    - **Role**: The "Digital Bouncer" that prevents scrapers and bots from stealing your content or spiking your AI costs.

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4. The AI Factory "Brain"

- **Amazon Bedrock**: The "Lead Editor" that uses LLMs to generate your technical repair manuals from video transcripts.
- **Amazon Rekognition**: The "Safety Inspector" that scans footage to automatically tag tools (e.g., "Use Wrench Here") and identify safety hazards.
- **Amazon Transcribe**: The "Secretary" that converts your garage audio/speech into text for the manuals.
- **Amazon Translate**: The "Globalizer" that converts your manuals into dozens of languages for your international student base.
- **Amazon Polly**: The "Narrator" that turns text manuals into high-quality audio voiceovers.

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5. The Interactive & Analytical Layer

- **AWS AppSync**
    - **Role**: Provides real-time GraphQL subscriptions so student progress bars and notifications update instantly without a page refresh.
- **Amazon EventBridge**
    - **Role**: The "Central Nervous System" that triggers the next step (like the AI Factory) the moment a video finishes transcoding.
- **Amazon Kinesis Data Firehose**
    - **Role**: The "Data Vacuum" that captures every "Play," "Pause," and "Seek" event for business growth analysis.